Estimation Theory

Sequential Importance Sampling

Bootstrap Filtering

Condensation Algorithm

Sequential Monte Carlo Filtering

Interacting Particle Approximation

Monte Carlo Filtering

Survival of Fittest

Particle Filter

Bayesian State Estimation

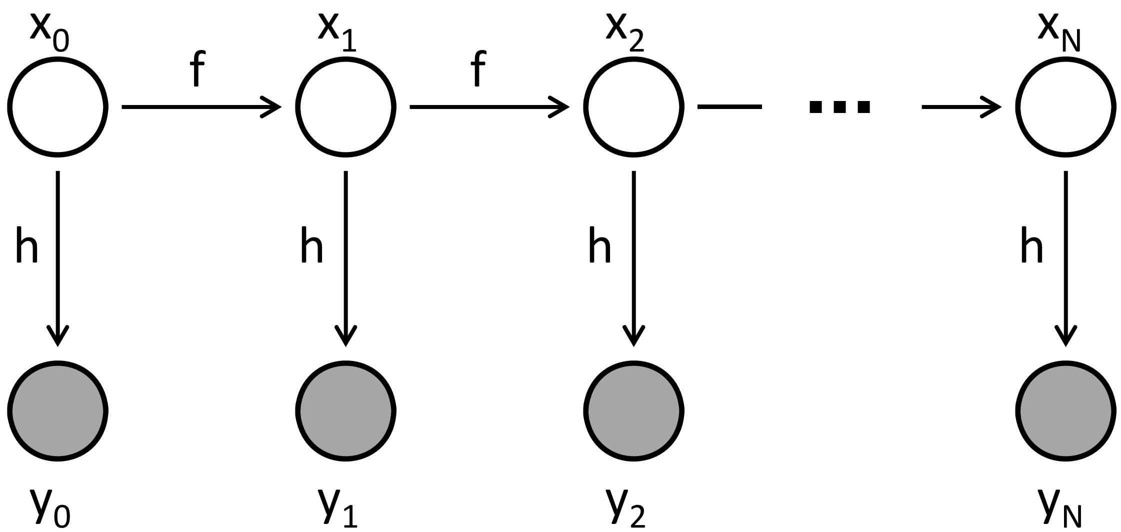
Nonlinear System

Discrete-time dynamic and measurement model

$$\boxed{\begin{aligned} x_{t+1} &= f(x_t, w_t) \\ y_t &= h(x_t, v_t) \end{aligned}}$$

- x_t: state at time t
- y_t: observation at time t
- w_t, v_t: white noise processes

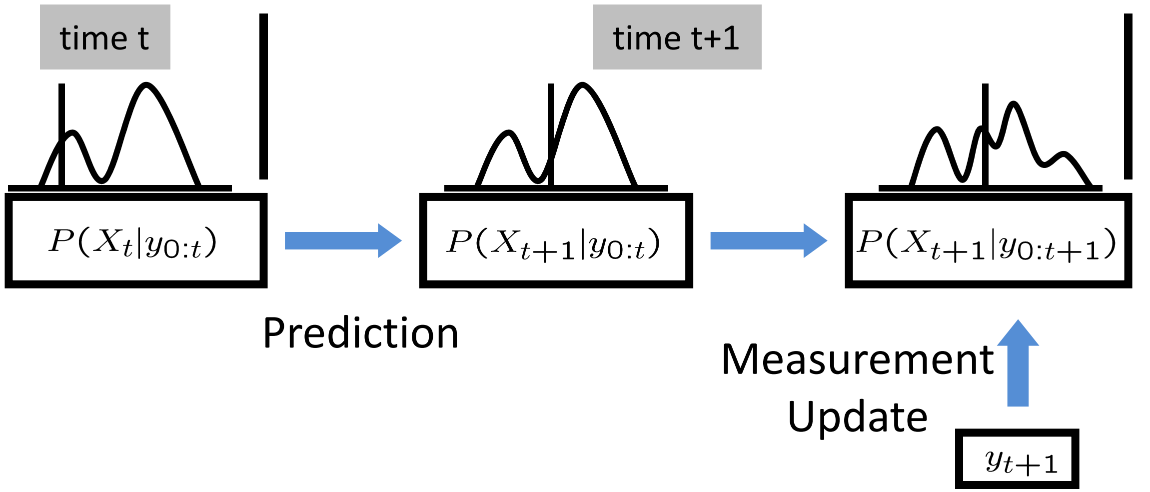
Graphical model:



Bayesian Filtering

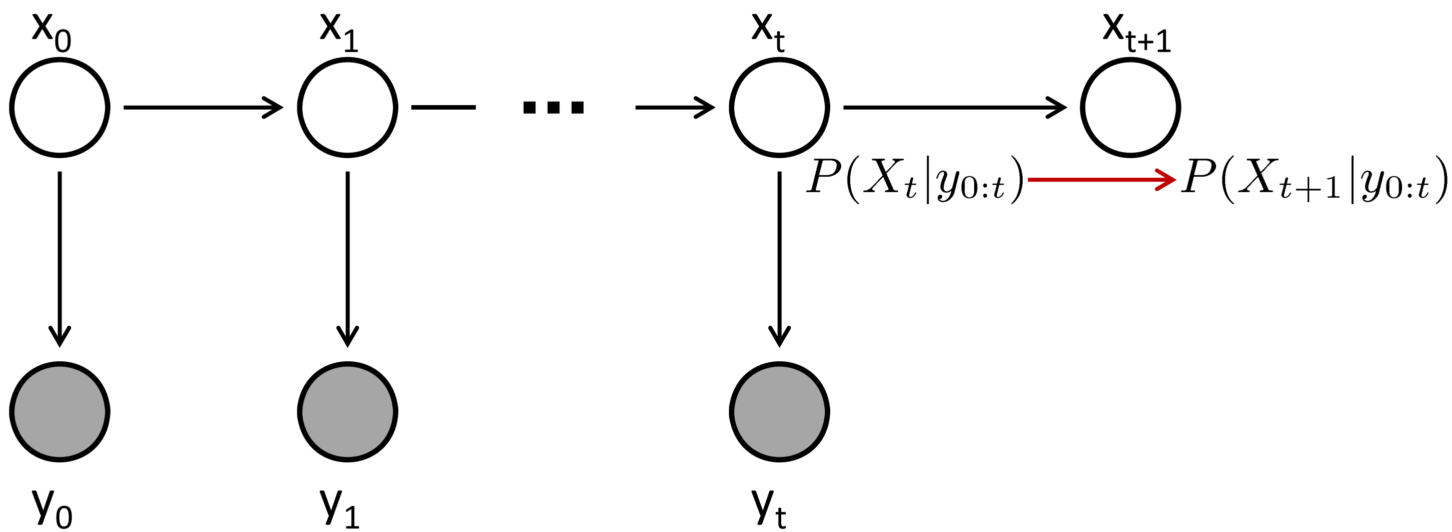
- Compute the posterior $P(X_t|y_{0:t})$ for each t, where $y_{0:t} = \{y_0, y_1, \ldots, y_t\}$.
- It is assumed that the prior $P(X_0)$ is known.
- Bayesian filtering is done in two steps: (1) prediction and (2) measurement update.
- Bayesian filtering algorithm

  1. Compute $P(X_{t+1}|y_{0:t})$ from $P(X_t|y_{0:t})$ (prediction)
  2. New observation y_{t+1}
  3. Compute $P(X_{t+1}|y_{0:t+1})$ (measurement update)
  4. Go to the next time



Examples of Bayesian filtering:
- Hidden Markov model,
- Kalman filters,
- Particle filters,
- …

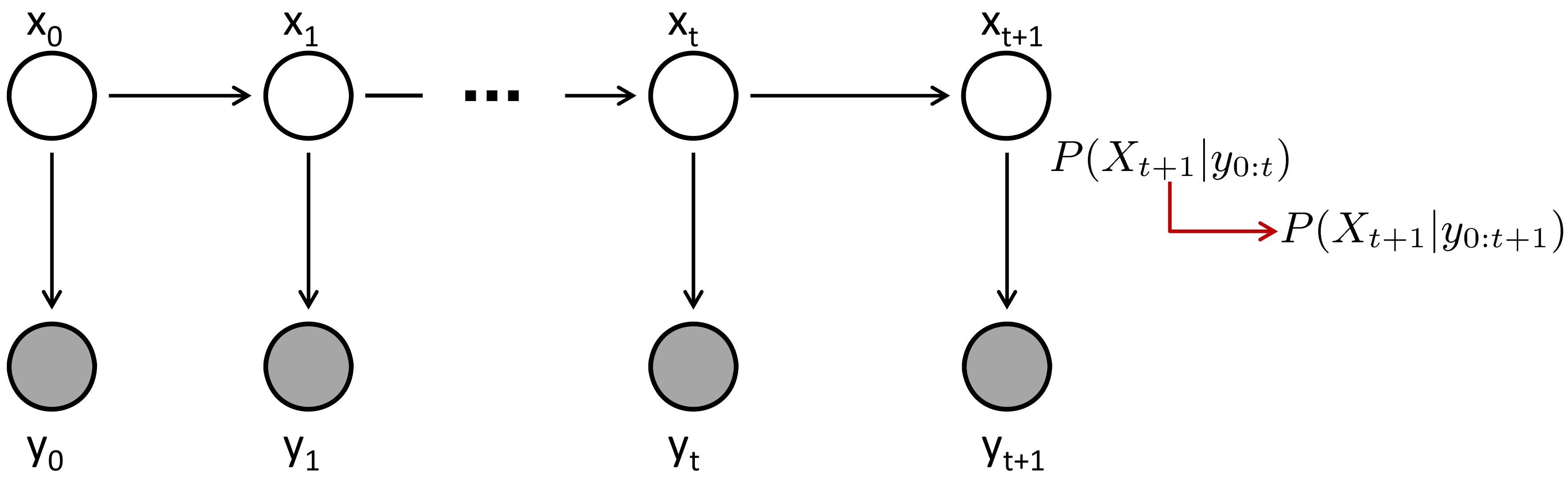
Dynamic Update



- $P(X_t|y_{0:t})$ is available from the previous time step.
- Prediction (or dynamic update):

$$\begin{aligned} P(X_{t+1}|y_{0:t}) &= \int P(X_{t+1}|x_t, y_{0:t}) P(x_t|y_{0:t}) dx_t \\ &= \int P(X_{t+1}|x_t) P(x_t|y_{0:t}) dx_t \end{aligned}$$

Measurement Update

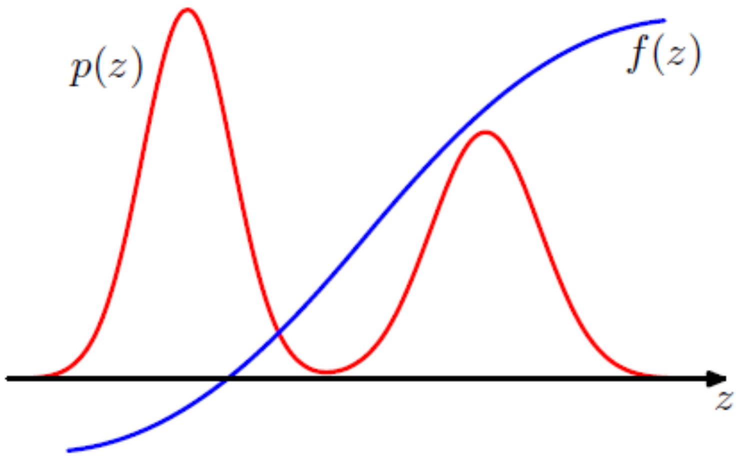


- New measurement y_{t+1}.
- Measurement update (Bayes rule):

$$
\begin{aligned}
P(X_{t+1}|y_{0:t+1}) &= \frac{P(y_{t+1}|X_{t+1}, y_{0:t})P(X_{t+1}|y_{0:t})}{\int P(y_{t+1}|x_{t+1}, y_{0:t})P(x_{t+1}|y_{0:t})dx_{t+1}} \\
&= \frac{P(y_{t+1}|X_{t+1})P(X_{t+1}|y_{0:t})}{\int P(y_{t+1}|x_{t+1})P(x_{t+1}|y_{0:t})dx_{t+1}}
\end{aligned}
$$

Sampling

Sampling



We want to compute the expected value of a function over a complex distribution

$$\mathbb{E}[f] = \int f(\mathbf{z})p(\mathbf{z})\,\mathrm{d}\mathbf{z}$$

But such expectations are too complex to be evaluated exactly using analytical techniques.

Approximate by samples $\{z^{(l)}\}$, where $z^{(l)}$ are drawn independently from p(z)

$$\widehat{f} = \frac{1}{L}\sum_{l=1}^{L} f(\mathbf{z}^{(l)})$$

$$\mathbb{E}[\widehat{f}] = \mathbb{E}[f]$$

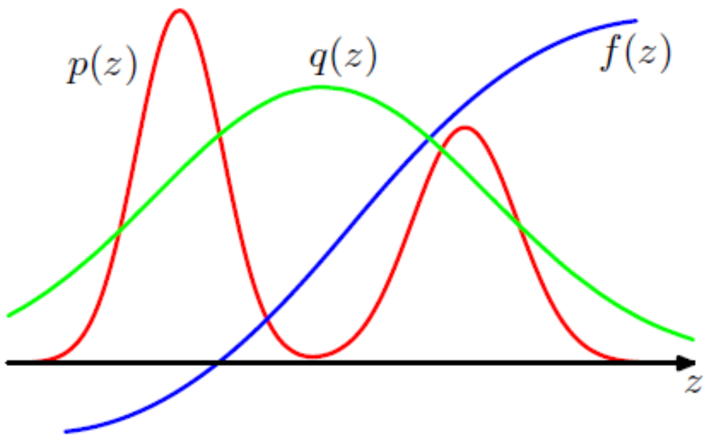
$$\mathrm{var}[\widehat{f}] = \frac{1}{L}\mathbb{E}\left[(f - \mathbb{E}[f])^2\right]$$

Importance Sampling

$$\mathbb{E}[f] = \int f(\mathbf{z})p(\mathbf{z})\,\mathrm{d}\mathbf{z} = \int f(\mathbf{z})\frac{p(\mathbf{z})}{q(\mathbf{z})}q(\mathbf{z})\,\mathrm{d}\mathbf{z} \simeq \frac{1}{L}\sum_{l=1}^{L}\frac{p(\mathbf{z}^{(l)})}{q(\mathbf{z}^{(l)})}f(\mathbf{z}^{(l)})$$

$\{\mathbf{z}^{(l)}\}$ drawn from $q(\mathbf{z})$

proposal distribution
(easy to sample from)



- $p(\mathbf{z}) = \frac{\tilde{p}(\mathbf{z})}{Z_p}$
- $\tilde{p}(\mathbf{z})$ can be easily evaluated
- Z_p is unknown
- $q(\mathbf{z}) = \frac{\tilde{q}(\mathbf{z})}{Z_q}$

$$\mathbb{E}[f] = \int f(\mathbf{z})p(\mathbf{z})\,\mathrm{d}\mathbf{z} = \frac{Z_q}{Z_p}\int f(\mathbf{z})\frac{\widetilde{p}(\mathbf{z})}{\widetilde{q}(\mathbf{z})}q(\mathbf{z})\,\mathrm{d}\mathbf{z} \simeq \frac{Z_q}{Z_p}\frac{1}{L}\sum_{l=1}^{L}\widetilde{r}_l f(\mathbf{z}^{(l)}).$$

$$\tilde{r}_l = \frac{\tilde{p}(\mathbf{z}^{(l)})}{\tilde{q}(\mathbf{z}^{(l)})}$$

$$\frac{Z_p}{Z_q} = \frac{1}{Z_q}\int \widetilde{p}(\mathbf{z})\,\mathrm{d}\mathbf{z} = \int \frac{\widetilde{p}(\mathbf{z})}{\widetilde{q}(\mathbf{z})}q(\mathbf{z})\,\mathrm{d}\mathbf{z} \simeq \frac{1}{L}\sum_{l=1}^{L}\widetilde{r}_l$$

$$\boxed{\mathbb{E}[f] \simeq \sum_{l=1}^{L} w_l f(\mathbf{z}^{(l)}) \qquad w_l = \frac{\widetilde{r}_l}{\sum_m \widetilde{r}_m} = \frac{\widetilde{p}(\mathbf{z}^{(l)})/q(\mathbf{z}^{(l)})}{\sum_m \widetilde{p}(\mathbf{z}^{(m)})/q(\mathbf{z}^{(m)})}}$$

Sampling-Importance-Resampling (SIR)

1. Draw L samples $\mathbf{z}^{(1)}, \ldots \mathbf{z}^{(L)}$ from $q(\mathbf{z})$
2. Compute weights $w_1, \ldots, w_L$
3. Draw another set of L samples from $\{\mathbf{z}^{(1)}, \ldots \mathbf{z}^{(L)}\}$ with probability $(w_1, \ldots, w_L)$

$$w_l = \frac{\widetilde{r}_l}{\sum_m \widetilde{r}_m} = \frac{\widetilde{p}(\mathbf{z}^{(l)})/q(\mathbf{z}^{(l)})}{\sum_m \widetilde{p}(\mathbf{z}^{(m)})/q(\mathbf{z}^{(m)})}$$

The resulting L samples are approximately distributed according to $p(\mathbf{z})$.

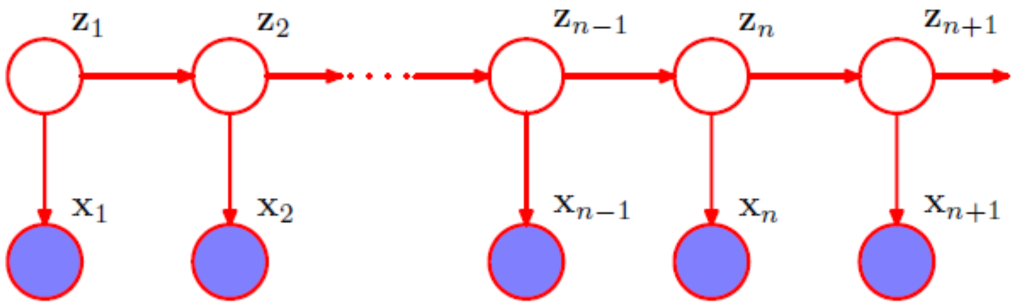
$$\begin{aligned} p(z \leqslant a) &= \sum_{l:z^{(l)} \leqslant a} w_l \\ &= \frac{\sum_l I(z^{(l)} \leqslant a)\widetilde{p}(z^{(l)})/q(z^{(l)})}{\sum_l \widetilde{p}(z^{(l)})/q(z^{(l)})} \end{aligned}$$

$(L \to \infty)$

$$\begin{aligned} p(z \leqslant a) &= \frac{\int I(z \leqslant a)\,\{\widetilde{p}(z)/q(z)\}\,q(z)\,\mathrm{d}z}{\int \{\widetilde{p}(z)/q(z)\}\,q(z)\,\mathrm{d}z} \\ &= \frac{\int I(z \leqslant a)\widetilde{p}(z)\,\mathrm{d}z}{\int \widetilde{p}(z)\,\mathrm{d}z} \\ &= \int I(z \leqslant a)p(z)\,\mathrm{d}z \end{aligned}$$

Particle Filtering

Particle Filter



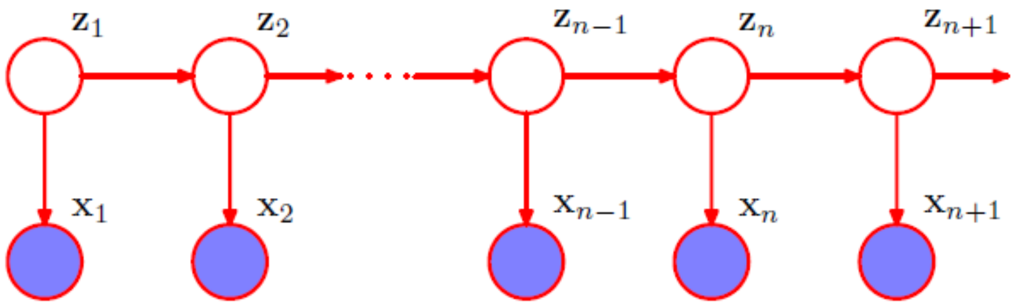
$$
\begin{aligned}
\mathbb{E}[f(\mathbf{z}_n)] &= \int f(\mathbf{z}_n) p(\mathbf{z}_n|\mathbf{X}_n)\,\mathrm{d}\mathbf{z}_n \\
&= \int f(\mathbf{z}_n) p(\mathbf{z}_n|\mathbf{x}_n, \mathbf{X}_{n-1})\,\mathrm{d}\mathbf{z}_n \\
&= \frac{\int f(\mathbf{z}_n) p(\mathbf{x}_n|\mathbf{z}_n) p(\mathbf{z}_n|\mathbf{X}_{n-1})\,\mathrm{d}\mathbf{z}_n}{\int p(\mathbf{x}_n|\mathbf{z}_n) p(\mathbf{z}_n|\mathbf{X}_{n-1})\,\mathrm{d}\mathbf{z}_n} \\
&\simeq \sum_{l=1}^{L} w_n^{(l)} f(\mathbf{z}_n^{(l)})
\end{aligned}
$$

- z_n: state at time n
- $\mathbf{X}_n = (x_1, \ldots, x_n)$: measurements
- $z_n^{(l)}$: a sample from $p(z_n|\mathbf{X}_{n-1})$
- $w_n^{(l)}$: sampling weight

$$
w_n^{(l)} = \frac{p(\mathbf{x}_n|\mathbf{z}_n^{(l)})}{\sum_{m=1}^{L} p(\mathbf{x}_n|\mathbf{z}_n^{(m)})}
$$

measurement update

Particle Filter



dynamic update

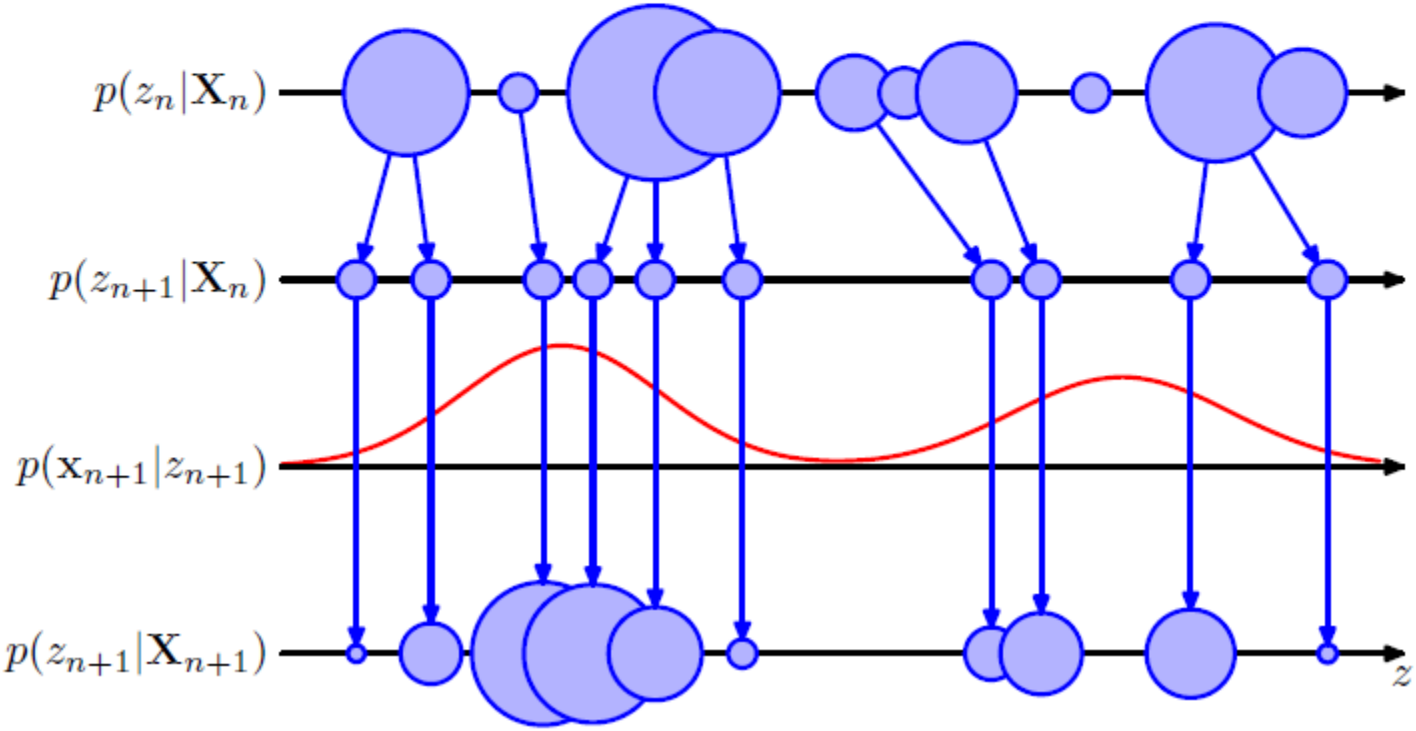
Sample from $p(\mathbf{z}_{n+1}|\mathbf{X}_n)$

$$\begin{aligned}
p(\mathbf{z}_{n+1}|\mathbf{X}_n) &= \int p(\mathbf{z}_{n+1}|\mathbf{z}_n, \mathbf{X}_n) p(\mathbf{z}_n|\mathbf{X}_n)\, \mathrm{d}\mathbf{z}_n \\
&= \int p(\mathbf{z}_{n+1}|\mathbf{z}_n) p(\mathbf{z}_n|\mathbf{X}_n)\, \mathrm{d}\mathbf{z}_n \\
&= \int p(\mathbf{z}_{n+1}|\mathbf{z}_n) p(\mathbf{z}_n|\mathbf{x}_n, \mathbf{X}_{n-1})\, \mathrm{d}\mathbf{z}_n \\
&= \frac{\int p(\mathbf{z}_{n+1}|\mathbf{z}_n) p(\mathbf{x}_n|\mathbf{z}_n) p(\mathbf{z}_n|\mathbf{X}_{n-1})\, \mathrm{d}\mathbf{z}_n}{\int p(\mathbf{x}_n|\mathbf{z}_n) p(\mathbf{z}_n|\mathbf{X}_{n-1})\, \mathrm{d}\mathbf{z}_n} \\
&= \sum_l w_n^{(l)} p(\mathbf{z}_{n+1}|\mathbf{z}_n^{(l)})
\end{aligned}$$

Particle Filter: Summary

Dynamic Update

Measurement Update



- At time step n, the posterior distribution $p(\mathbf{z}_n|\mathbf{X}_n)$ is represented by samples $\{\mathbf{z}_n^{(l)}\}$ with corresponding weights $\{w_n^{(l)}\}$.
- For $n+1$, draw L samples from the mixture distribution $\sum_l w_n^{(l)} p(\mathbf{z}_{n+1}|\mathbf{z}_n)$ (dynamic update).
- Each sample is weighted using the new measurement $\mathbf{x}_{n+1}$ as $w_{n+1}^{(l)} \propto p(\mathbf{x}_{n+1}|\mathbf{z}_{n+1}^{(l)})$ (measurement update).

Particle Filter: Implementation Issues

- **Sample impoverishment**: when $p(y_k|x_k)$ does not overlap with $p(x_k|y_{k-1})$, only a few particles will survive. Eventually, all particles will collapse to the same value. The problem is more serious for higher dimensional problems.

- **Roughening**: add a random noise to each particle after resampling.

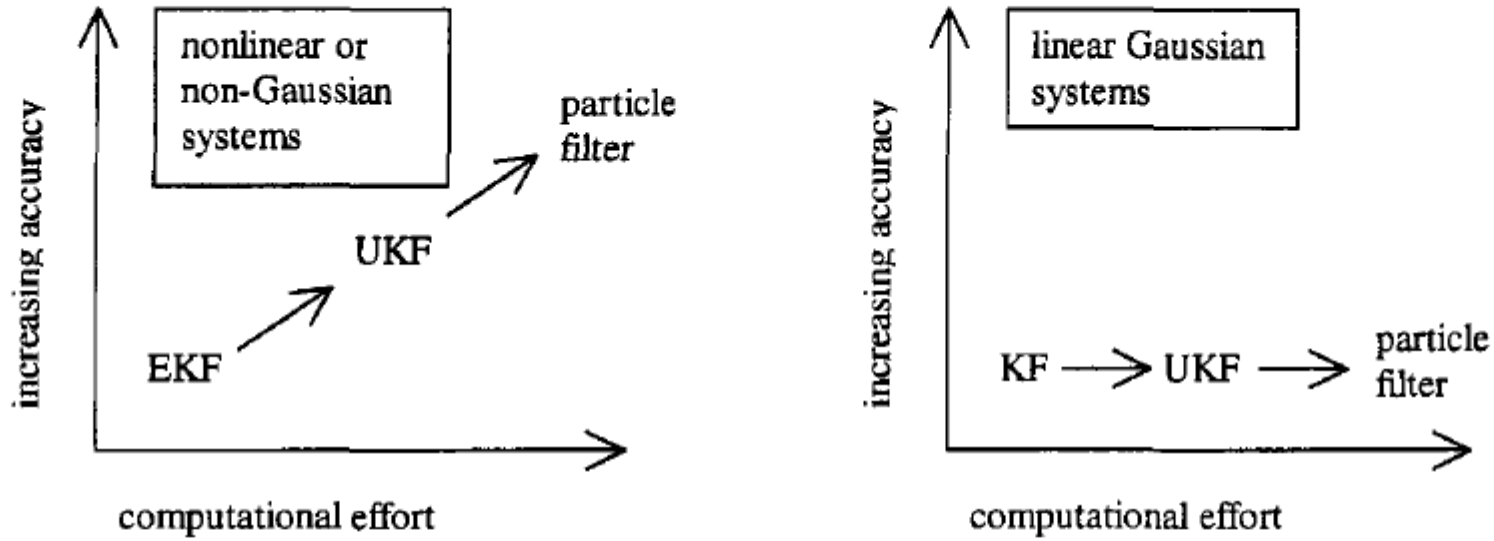
- **Regularized particle filtering**: continuous approximation of $p(x_k|y_k)$ (kernel regression).

- **Markov chain Monte Carlo (MCMC) resampling**

- **Auxiliary particle filtering**: even out probabilities of the *a priori* particles. One example:
$$\tilde{q}_i = \frac{(\alpha - 1)q_i + \bar{q}}{\alpha},$$
where $\bar{q}$ is the sample mean of all q_i and $\alpha \in [1, \infty]$.

- **Particle filtering combined with other filters**: e.g., extended Kalman particle filter.

Summary


nonlinear or non-Gaussian systems
particle filter
UKF
EKF
increasing accuracy
computational effort
linear Gaussian systems
KF
UKF
particle filter
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